Midterm 2, COSC330/PHYS306

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1 Adders

[12 points]. A 4-bit ripple-carry adder adds $A = 1011$ and $B = 0110$ with an initial carry-in of $C_{\text{in}} = 0$.

1. Determine the 4-bit sum and the final carry-out.

2. Verify the result using ordinary binary addition.

3. Suppose the worst-case carry propagation delay through each full-adder stage is 7 ns. Estimate the worst-case total carry delay for the 4-bit adder.

4. If the addition must finish within one clock period, estimate the maximum clock frequency.

2 Comparators, Decoders, and Encoders

[12 points]. Answer the following questions about comparators, decoders, and encoders.

1. A 4-bit comparator receives $A = 1101$ and $B = 1011$. Which output is HIGH: $A > B$, $A = B$, or $A < B$?

2. Explain how four XNOR gates can be used as part of a 4-bit equality comparator. What additional gate is required? Draw the comparator using basic gates.

3. A 4-to-16 decoder receives $A_3A_2A_1A_0 = 1010$. Which output line should be asserted?

4. A decimal-to-BCD encoder has only the decimal input line 7 asserted. What 4-bit output should appear? Draw the gates and connections that represent that segment of the encoder.

3 Multiplexers as Logic Functions

[12 points]. A logic function $Y(A, B, C)$ is specified by

A	B	C	Y
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

Use a 8-to-1 multiplexer to implement this function.

1. Treat A, B, and C as the data-select inputs, and for each combination, determine whether a HIGH or LOW signal must be connected to the corresponding data input. Draw the completed multiplexer block diagram, including the eight data-input connections, the three select inputs, and the output Y.

4 Latches and Flip-Flops

[14 points]. Answer the following questions about level-sensitive latches and edge-triggered flip-flops.

1. A gated D latch initially has $Q = 0$. While $EN = 1$, the input D changes according to

   $$0,\ 1,\ 0,\ 1.$$

   List the values of Q while the latch is enabled.

2. A positive-edge-triggered D flip-flop initially has $Q = 0$. Immediately before four successive active clock edges, the values of D are

   $$1,\ 1,\ 0,\ 1.$$

   List the value of Q immediately after each edge.

3. A JK flip-flop initially has $Q = 1$. At three successive active clock edges the inputs are

   $$(J, K) = (0, 1), \quad (1, 0), \quad (1, 1).$$

   Determine Q after each clock edge.

4. Briefly explain the difference between a latch and an edge-triggered flip-flop.

5 Shift Registers

[12 points]. A 4-bit serial-in/serial-out shift register is initially $Q_3Q_2Q_1Q_0 = 0000$. On each active clock edge, a new bit enters Q_3, and the existing bits shift one position toward Q_0. The serial input sequence is 10110.

1. Complete the register state after each clock edge:

Clock edge	Serial input	$Q_3Q_2Q_1Q_0$
0	–	0000
1	1	
2	0	
3	1	
4	1	
5	0	

2. At what clock edge does the first input bit reach Q_0?

3. If the clock frequency is 2 MHz, how much time passes before that first bit reaches Q_0?

6 Counters

[12 points]. Answer the following questions about counters.

1. A 4-bit binary up-counter starts at 0000. What state does it contain after 11 clock pulses?

2. A 4-bit ripple counter is driven by a 200 kHz clock. How much time is required to count from 0000 to 1111?

3. Explain why a ripple counter is called *asynchronous*.

4. State one advantage of a synchronous counter over an asynchronous counter.

7 ADC, Sampling, and DAC Errors

[14 points]. Answer the following questions about sampling, quantization, and conversion errors.

1. An ADC samples once every

   $$\Delta t = 50\,\mu\text{s}.$$

   Determine the sampling frequency f_s and the Nyquist frequency f_N.

2. A 12-bit ADC covers a nominal range of 0 to 2.0 V. Estimate the voltage resolution. That is, what is the change in input voltage if the ADC output increases by one count?

3. A sinusoid of frequency 7.5 kHz is sampled at 20 kHz. How many samples per period will the ADC capture?

4. Characterize the following conversion errors as constant one of the following: voltage offsets, incorrect gain, or non-linear calibration:

   (a) every output is shifted up by the same voltage

   (b) the slope is consistently too small

   (c) the transfer curve bends away from a straight line

   (d) the slope is consistently too large

8 Design Problem: Digital Sawtooth Synthesizer

[12 points]. A simple digital synthesizer can produce a sawtooth waveform by connecting the output of a binary counter to a digital-to-analog converter (DAC). Consider a 4-bit binary up-counter whose outputs $Q_3Q_2Q_1Q_0$ are connected directly to the inputs of an ideal 4-bit DAC. The DAC produces 0 V for input 0000 and 3.0 V for input 1111, with equally spaced voltage levels between them.

1. Draw a block diagram showing the clock driving the 4-bit counter, the four counter outputs connected to the DAC, and the analog output voltage V_{out}.

2. Starting from 0000, list the sequence of DAC input codes for one complete counter cycle. Explain why the DAC output forms a staircase approximation to a sawtooth waveform, including what happens when the counter rolls over from 1111 to 0000.

3. Determine the DAC voltage step, ΔV, corresponding to an increase of one binary count. Then determine V_{out} for the counter states 0101 and 1100.

4. Suppose the counter is driven by an 8.0 kHz clock.

   (a) What is the period of one complete sawtooth waveform?

   (b) What is the fundamental repetition frequency of the sawtooth?

   (c) What clock frequency would be required to produce a sawtooth with fundamental frequency 440 Hz?